

RESEARCH STATEMENT

My research interests center on program synthesis: how specifications, examples, and symbolic constraints can guide the construction of correct programs. I am especially interested in formal methods that connect the syntax of programs and specifications with their semantic meaning. Before moving into research, I specialized in air-gapped software at Google Cloud.

EDUCATION

Ashoka University
Bachelor of Science (Honours) in Computer Science
Major CGPA: 3.74/4

Sonipat, Haryana
Aug 2018 - Apr 2022

RELEVANT COURSEWORK

Discrete Mathematics; Algorithm Design and Analysis; Linear Algebra; Calculus; Abstract Algebra; Probability Theory; Programming Language Design and Implementation; Operating Systems; Computer Networks; Computer Security and Privacy.

EMPLOYMENT

Ashoka University
Research Associate · Program Synthesis

Sonipat, Haryana
Jun 2026 - Present

- Working with [Prof. Aalok Thakkar](#) on program synthesis, symbolic logic, formal methods, and formal reasoning.
- Studying questions at the boundary of syntax and semantics, including how symbolic representations and specifications acquire computational meaning in program synthesis and formal reasoning workflows.

Firebolt
Software Engineer · Self-hosted Platform

Bangalore, India
Nov 2025 - Mar 2026

- Joined the founding engineering team building Firebolt's first self-hosted platform for isolated, air-gapped enterprise deployments, including feature flags and Temporal workflow orchestration.

Google
Software Engineer · Google Cloud, Distributed Cloud Team

Bangalore, India
Mar 2024 - Sept 2025

- Built software for Google Distributed Cloud, bringing Google Cloud services into air-gapped, sovereign, and regulated environments through distributed systems that operated without public-network dependencies.
- Designed a Harbor-based artifact registry for a 40kg portable data center, adding Kubernetes self-healing, node-local fallback mirrors, and image signature verification to support offline bootstrap and operation.
- Built APIs and observability tooling for Prometheus/Grafana monitoring, covering system health, storage, image-pull latency, error rates, alerting, and dashboards for internal teams and deployment operators.
- Mentored junior engineers and hosted a summer intern who later joined full time.

Google
Strategic Cloud Engineer · Google Cloud

Bangalore, India
Jul 2022 - Mar 2024

- Helped enterprises become cloud-native on Google Cloud, combining technical consulting, hands-on implementation, and architecture guidance across cloud infrastructure projects.
- Researched and prototyped Data Insights, an LLM-powered natural-language layer for tabular data discovery, testing generated questions, SQL guardrails, accuracy, and latency for cold-start exploration of unfamiliar datasets; the capability later became part of a Google Cloud data management product.
- Built reusable cloud and data tooling, including Terraform modules for Kubernetes clusters and VPCs alongside open-source Dataproc templates for Spark/Hadoop ETL.

TEACHING

AshokaX
Teaching Assistant · Machine Learning

Online
Apr 2022 - Jun 2022

- Assisted [Prof. Debayan Gupta](#) in teaching machine learning to high school students.

Ashoka University
Teaching Assistant · Computer Organization and Systems

Sonipat, Haryana
Aug 2021 - Dec 2021

- Assisted [Prof. Manu Awasthi](#) with Computer Organization and Systems in Ashoka's Computer Science department.